Visualization of Single Perceptron

$$X = \begin{bmatrix} x_1; \\ x_2; \\ \dots \\ x_n \end{bmatrix} \rightarrow X = \begin{bmatrix} 1; \\ x_1; \\ x_2; \\ \dots \\ x_n \end{bmatrix}$$

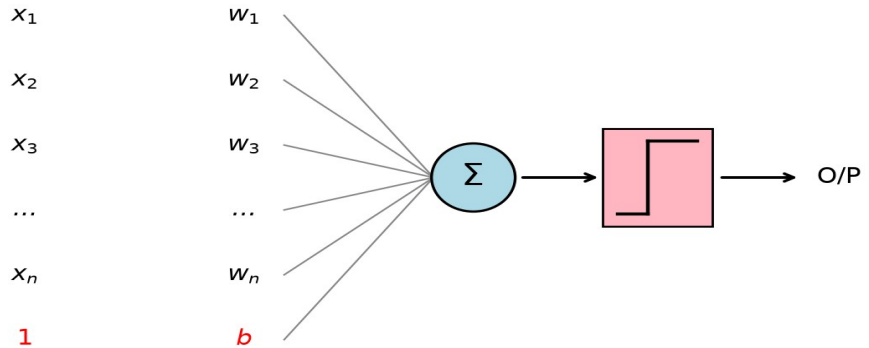
Augmented Vector

$$W = \begin{bmatrix} b; \\ w_1; \\ w_2; \\ \dots \\ w_n \end{bmatrix}$$

Model Weights

$$d(X) = W^T X = w_1x_1 + w_2x_2 + \dots\dots + w_nx_n + b*1$$

}
>0 ⇒ Class I
<0 ⇒ Class II
=0 ⇒ Undefined Class



Binary Classification (2 Classes)